

Name_____

Statistics 1040: Introduction to Statistics

Section 12

Test 2 – November 9, 2000

Version 1

75 minutes; closed book, closed notes, calculators and provided formula sheet allowed. 15 questions, 7 points each (max score is 105). For numeric questions, show your work and circle your answers. For essay questions, answer the questions in complete sentences. Good luck!

1. I propose to you a new game. You'll roll a fair (standard, six-sided) die twice. If you roll at least one 6, you win and I'll pay you \$2. If both rolls are each less than 6, I win, and you pay me \$1. What is the chance you will win on the first play of this game?
2. Continuing from problem 1, suppose we play this game 100 times. Describe a box model for your winnings at the end of this series of plays. Be complete!
3. Use the box model from problem 2 to find the chance that you will be ahead (that is, have winnings of more than \$0) after the 100 plays.

4. We draw two cards (without replacement) from a well-shuffled, standard, 52-card deck (4 suits of 13 cards each). Are the events “the first card is a spade” and “the first card is an ace” independent, mutually exclusive, both, or neither? Justify your answer.
5. We wish to estimate the percentage of Utahns in various parts of the state who support hosting the 2002 Olympics. Suppose that we are happy with the accuracy of a survey of 500 individuals in Logan. Salt Lake City is, of course, much larger than Logan, so to get equivalent accuracy there, we should take a sample which is: (pick one and explain: much smaller than 500, about 500, much larger than 500).
6. We draw 6 times with replacement from the box

1	2	3	4	5
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. What is the chance we will draw exactly 5 odd numbers?
7. True or false and explain: As we make more and more draws from the box in problem 6, the data histogram of the draws should look more and more like the normal curve.

8. The April, 1999, issue of *Professional Psychology: Research and Practice* includes an article, “Sexuality on the Internet from Sexual Exploration to Pathological Expression.” This article presents the results of a survey which visitors to the MSNBC web site were invited to take. The survey consisted of 59 questions including demographic information such as age, gender, and occupational status, as well as many questions relating to use of sexually oriented internet resources, such as adult web sites, newsgroups, and chat rooms. The researchers used the results of 9,177 such completed surveys. They report that 8% of respondents spend 11 or more hours per week engaging in on-line sexual activities. Assuming that 734 of the surveys indicated such behavior, and that the surveys represent a simple random sample of all internet users, calculate a 95% confidence interval for the percentage of all internet users who spend 11 or more hours weekly in such activities.
9. Explain, without using the word “confidence”, what the 95% in the interval in problem 8 represents.
10. Is the assumption in problem 8 that the survey respondents represent a simple random sample of all internet users a good one? Explain why or why not, and what implications your answer has for the results of problem 8.
11. The manager of a 100-unit condominium complex interviews the 100 condo owners in the complex and finds 12 plan to sell their condos in the next year. He estimates 12% plan to sell. By how much is his estimate likely to be off? Justify your answer.

12. A company with 1000 employees takes a simple random sample of 100 employees, and checks their records to find the number of days lost due to illness in the past year. The 100 numbers have an average of 7.0 days and an SD of 2.5 days. What is our best estimate for the average number of days lost for all employees of the company? By about how much should we expect to be off?
13. Five cards are drawn without replacement from a well-shuffled standard, 52-card deck (4 suits - clubs, diamonds, hearts, and spades - with 13 cards each). What is the chance that all are hearts?
14. The 7-Eleven stores in New York City let their customers answer a weekly survey by choosing to have their coffee in a cup labeled yes or no. A January 1990 survey asked if Congress deserved a pay raise, illustrating the question with a picture of a fat congressman lighting a cigar with a \$100 bill. 71 percent of the 682,285 persons served chose a no cup. Identify at least two problems with this survey.
15. A fair coin is flipped a number of times. Which is more likely: 45% to 55% heads in 100 tosses, or 45% to 55% heads in 400 tosses?